

LABWORK 4

This lab aims to help you practice loops and basic algorithms by solving **real-life inspired problems** using **functions** in C. You will implement 10 functions, each focusing on a different type of algorithmic thinking — calculation, iteration, and pattern generation.

Function 1: sum_first_n_numbers

Return Type: `int` **Parameters:** `int n` **Description:** Calculate the sum of the numbers from 1 to n.

Example: Input: 5 Output: 15 (1+2+3+4+5)

Function 2: print_number_square (5 pts)

Return Type: `void` **Parameters:** `int n` **Description:** Print a square of numbers from 1 to n, where each row is filled with the same numbers.

Example: Input: 4 Output: 1 1 1 1 2 2 2 2

Function 3: print_number_sequence (5 pts)

Return Type: `void`

Parameters: `int n`

Description: Prints numbers from 1 to n in a single line separated by spaces.

Example:

Input: 5 Output: 1 2 3 4 5

Function 4: print_multiples_of_three

Return Type: `void` **Parameters:** `int n` **Description:** Print the multiples of 3 from 1 to n.

Example: Input: 10 Output: 3 6 9

Function 5: print_reverse_numbers

Return Type: `void` **Parameters:** `int n` **Description:** Print the numbers from 1 to n in reverse order.

Example: Input: 5 Output: 5 4 3 2 1

Function 6: print_temperature_pattern (10 pts)

Return Type: `void`

Parameters: `int days`

Description: Prints a pattern of temperature bars using `*`, where each line represents a day and the number of stars increases by 1.

Example:

For `days = 4`:

```
Day 1: *
Day 2: **
Day 3: ***
Day 4: ****
```

Function 7: `print_fuel_consumption_table` (10 pts)

Return Type: `void`

Parameters: `int liters`

Description: Prints the **distance covered** per liter of fuel (assume 15 km/liter) for 1 to the given `liters`.

Example:

```
For `liters = 5`:
1 liter = 15 km
2 liters = 30 km
3 liters = 45 km
4 liters = 60 km
5 liters = 75 km
```

Function 8: `print_hollow_square` (10 pts)

Return Type: `void` **Parameters:** `int size` **Description:** Print a square of size `size`, where the outer edges are `*` (stars), and the inner is filled with space characters.

Example: For `size = 5`:

```
*****
*  *
*  *
*  *
*  *
*****
```

Function 9: `print_bus_seat_pattern` (10 pts)

Return Type: void

Parameters: int rows

Description:

Prints the seating layout of a bus.

Each row has 4 seats labeled [A] [B] [C] [D].

Example:

For rows = 3:

```
[A] [B] [C] [D]
[A] [B] [C] [D]
[A] [B] [C] [D]
```

****Function 10: print_bus_schedule**

Return Type: void **Parameters:** int stops **Description:** Print the bus schedule for the given number of stops, where each stop is 5 minutes apart.

Example: Input: 3 stops Output: Stop 1: 5 minutes Stop 2: 10 minutes Stop 3: 15 minutes ... Stop n: (n-1)*5 minutes

Main Function (15 pts)

Function Name: main

Return Type: int

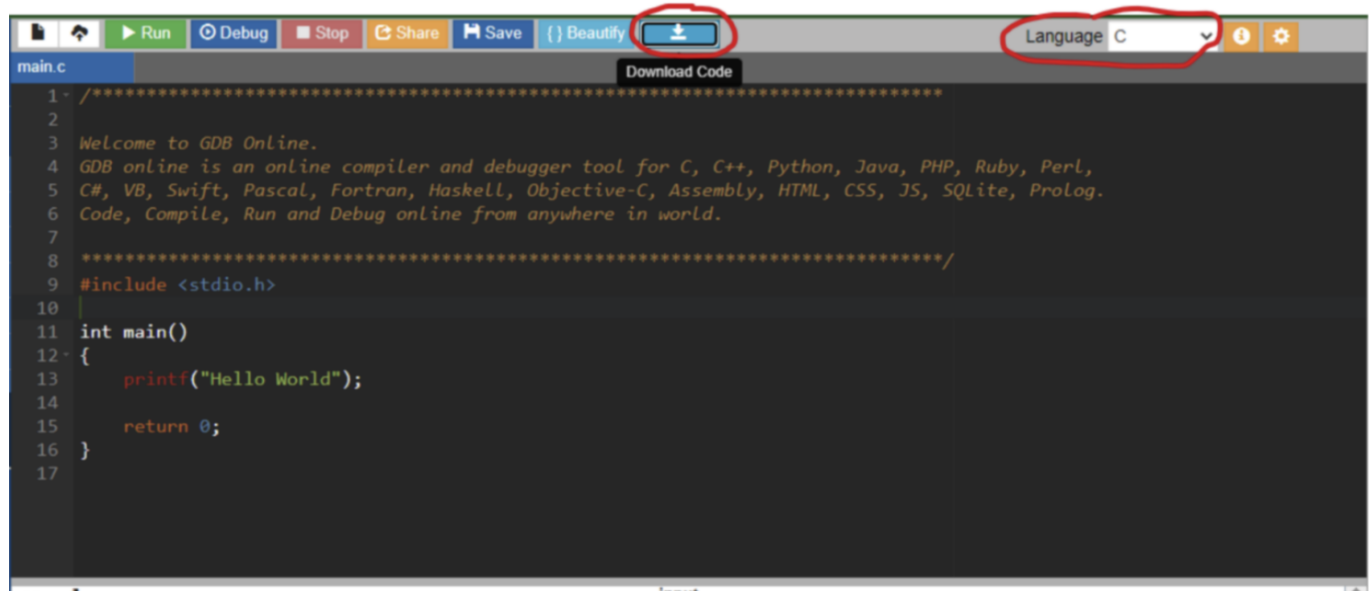
Parameters: none

Description:

- Ask the user for input arrays and numbers.
 - Call each of the 10 functions.
 - Print the results clearly and formatted.
 - Demonstrate proper input validation and loop usage.
-

Submission Format

- Submit your .c file named as:
name_surname.c
- Late submissions will not be accepted.
- Follow function naming and signature exactly.
- Clear screen and print statements are optional, not graded.



```
1  /******  
2  
3  Welcome to GDB OnLine.  
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,  
5  C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.  
6  Code, Compile, Run and Debug online from anywhere in world.  
7  
8  *****/  
9  #include <stdio.h>  
10  
11 int main()  
12 {  
13     printf("Hello World");  
14  
15     return 0;  
16 }  
17
```

Important Notes

- You are encouraged to modularize your code by creating functions where applicable.
- If you're unsure about the use of specific features or need assistance, refer to the course material or ask your instructor.